

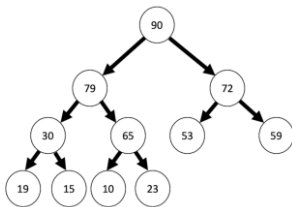
# Question 1

- (a) Show that the complete binary tree with  $n$  nodes has height at most  $O(\log n)$ .
- (b) If a complete binary tree has 127 nodes, calculate the maximum possible height of the tree.

## Question 2

(a) Consider the following max heap represented as a binary tree:

Show the process of removing the node with value 23 from the heap. Draw all intermediate trees after each step of rebalancing the heap.



(b) Given that a complete binary tree has a height of  $O(\log n)$ , justify why the time complexity of removing a node from a heap and restoring the heap property is  $O(\log n)$ .